Assignment 08

1. Implement the following function to find an element in an unsorted array, using linear search.

```
package in.cdac;
import java.util.Scanner;
public class LinearSearch {
	public static boolean findUsingLinearSearch(int[]
arr, int element) {
		for(int i=0; i<arr.length; i++) {
			if(arr[i] == element)  return true;
		}
		return false;
	}
	public static void main(String[] args) {
		Scanner scanner = new
Scanner(System.in);

		int[] arr = {1,2,3,4,5,6,7,8,9,10};
		System.out.println("Enter element to be
searched: ");
		int data = scanner.nextInt();
```

```
System.out.println(findUsingLinearSearch(arr,
data));

            scanner.close();
        }
}
```

Output:-

```
Console ×
<terminated> LinearSearch [Java Application] C:\Program Files\eclipse\plugins\org.eclipse
Enter element to be searched:

4
true
```

2. Implement the following function to find an element in a sorted array, using binary search. Hint: Sort the array using the sorting algorithm implemented below.

```
package in.cdac;
import java.util.Scanner;
public class BinarySearch {
    public static boolean
findUsingBinarySearch(int[] arr, int element) {
        int start = 0;
        int end = arr.length-1;

        while(start<=end) {
            int mid = start+(end-start)/2;
            if(arr[mid] == element) return true;

            if(arr[mid]<element) {
                start=mid+1;
            }
            if(arr[mid]>element) {
                end=mid-1;
        }
        }

        return false;
```

```
    }
    public static void main(String[] args) {
        Scanner scanner = new
Scanner(System.in);

        int[] arr = {1,2,3,4,5,6,7,8,9,10};
        System.out.println("Enter element to be
searched: ");
        int data = scanner.nextInt();


System.out.println(findUsingBinarySearch(arr,
data));

        scanner.close();
    }
}
```

Output:-

```
Console
<terminated> BinarySearch [Java Application] C:\Program Files\eclipse\plugins\org.eclipse.ju
Enter element to be searched:
5
true
```

3. Implement the following function to sort the given array in descending order, using Bubble Sort. Use the algorithm described in class to start from left side of array, to move smallest element towards the right end.

```
package in.cdac;
import java.util.Scanner;
public class BubbleSort {
    public static void bubbleSort(int[] arr) {
        for(int i=0; i<arr.length-1; i++) {
            for(int j=0; j<arr.length-i-1; j++) {
                if(arr[j] < arr[j+1]) {
                    int temp = arr[j+1];
                    arr[j+1] = arr[j];
                    arr[j] = temp;
                }
            }
        }
    }

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        int[] arr = {5,8,6,7,2,3,9,1};
        System.out.println("Original array: ");
```

```
            for(int i = 0; i<arr.length; i++) {
                System.out.print(arr[i]+" ");
            }
            System.out.println();
            bubbleSort(arr);
            System.out.println("Sorted array: ");

            for(int i = 0; i<arr.length; i++) {
                System.out.print(arr[i]+" ");
            }
//
System.out.println(findUsingBinarySearch(arr,
data));

            scanner.close();
        }
}
```

Output:

```
Console
<terminated> BubbleSort [Java Application] C:\Program Files\eclipse\plugins\org.eclipse.just
Original array:
5 8 6 7 2 3 9 1
Sorted array:
9 8 7 6 5 3 2 1
```

4. Implement the following function to sort the given array in descending order, using Insertion Sort

```
package in.cdac;
import java.util.Scanner;
public class InsertionSort {

    public static void insertionSort(int[] arr) {
        for(int i=1; i<arr.length; i++) {
            int current = arr[i];
            int j = i-1;

            while(j>=0 && current>arr[j]) {
                arr[j+1] = arr[j];
                j--;
            }
            arr[j+1] = current;
        }
    }
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        int[] arr = {5,8,6,7,2,3,9,1};
        System.out.println("Original array: ");
```

```
        for(int i = 0; i<arr.length; i++) {
            System.out.print(arr[i]+" ");
        }
        System.out.println();
        insertionSort(arr);
        System.out.println("Sorted array: ");

        for(int i = 0; i<arr.length; i++) {
            System.out.print(arr[i]+" ");
        }

        scanner.close();
    }
}
```

Output:

```
Console
<terminated> InsertionSort [Java Application] C:\Program Files\eclipse\plugins\org.eclipse.ju
Original array:
5 8 6 7 2 3 9 1
Sorted array:
9 8 7 6 5 3 2 1
```

5. Implement the following function to check if given array elements are sorted or not. Hint: Use logic used in bubble sort.

```
package in.cdac;
import java.util.Scanner;
public class IsSorted {

    public static void bubbleSort(int[] arr) {
        for(int i=0; i<arr.length-1; i++) {
            for(int j=0; j<arr.length-i-1; j++) {
                if(arr[j] > arr[j+1]) {
                    int temp = arr[j+1];
                    arr[j+1] = arr[j];
                    arr[j] = temp;
                }
            }
        }
    }

    public static boolean isArraySorted(int[] arr) {
        for(int i=0; i<arr.length-1; i++) {
            if(arr[i] > arr[i+1]) {
                return false;
            }
        }
```

```
        return true;
    }

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        int[] arr = {5,8,6,7,2,3,9,1};
        System.out.println("Original array: ");

        for(int i = 0; i<arr.length; i++) {
            System.out.print(arr[i]+" ");
        }
        System.out.println();
        System.out.println(isArraySorted(arr));
        bubbleSort(arr);
        System.out.println("Sorted array: ");

        for(int i = 0; i<arr.length; i++) {
            System.out.print(arr[i]+" ");
        }
        System.out.println(isArraySorted(arr));

        scanner.close();
    }
```

```
}
```

Output:-

```
Console
<terminated> IsSorted [Java Application] C:\Program Files\eclipse\plugins\org.eclipse.justj.
Original array:
5 8 6 7 2 3 9 1
false
Sorted array:
1 2 3 5 6 7 8 9 true
```